

02. Contraceptive Effectiveness and Method Families

Companion reading handout for simultaneous listening.

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A patient asks a question that sounds simple:

What is the best birth control?

If the resident answers with a brand name, the counseling has already become too shallow.

The right response is not, this method is best.

The right response is, best for what?

Best at preventing pregnancy? Best for heavy bleeding? Best without estrogen? Best without a pelvic procedure? Best for privacy? Best for future fertility? Best while taking enzyme-inducing medication? Best when pregnancy would be medically dangerous? Best for reducing some sexually transmitted infection risk? Best for a patient who knows she will not remember a daily pill?

That question, "best for what," is the foundation of contraceptive counseling.

The dominant model for this lecture is this:

Contraceptive effectiveness is not only pharmacology. It is biology, device design, and human behavior over time.

A method can fail because the biology is weak.

It can fail because the device is used incorrectly.

It can fail because it asks too much from a real person living a real life.

And sometimes the method does not fail at all. The counseling fails, because the method was effective in general but a poor fit for the patient's body, goals, or context.

The FDA Birth Control Guide is useful because it makes one part of this visible. It gives high-level information about method groups and estimates the number of pregnancies expected in the first year for each one hundred women using the method.

The key phrase is actual use.

The chart is not describing a perfect resident, a perfect patient, and a perfectly controlled trial world. It is trying to show what happens when methods are used in real life, including use that is not correct or not consistent.

That is why the chart is powerful.

It does not only rank products. It exposes where failure enters the system.

So before we climb the method ladder, make the first distinction:

Perfect use is what the method can do under ideal conditions.

Typical use is what happens when humans use it.

For a method that requires very little ongoing behavior, those two numbers tend to be closer.

For a method that depends on daily, weekly, monthly, or pericoital behavior, typical use becomes much more important.

This is not a moral judgment about patients.

It is system design.

A method that requires a correct action every day has three hundred sixty five possible adherence moments each year.

A method that requires correct use with every act of intercourse has a failure surface around intimacy, timing, availability, partner cooperation, substance use, discomfort, and negotiation.

A method that is placed once and then works for years has a different failure surface. Placement matters. Expulsion matters. Contraindications matter. But daily memory does not.

That is the first clinical pearl:

Every contraceptive method has a failure surface.

Your job is to identify where that surface is for this patient.

Start at the top of the FDA chart

The least pregnancy-prone group in the first year contains female sterilization, male sterilization, copper IUD or IUS, hormonal IUD or IUS, and the implantable rod. The chart places these in the less-than-one pregnancy per one hundred women in the first year group.

That number matters because it changes counseling when pregnancy would carry high risk.

If a patient has severe pulmonary hypertension, major cardiac disease, teratogenic medication exposure, or another condition where unintended pregnancy could be dangerous, then "reasonably effective" may not be good enough. The resident should think hard about methods that remove user dependence.

But do not flatten this top group.

Sterilization and long-acting reversible contraception are not the same thing.

Sterilization is highly effective because it is procedural and permanent. The patient does not have to remember a daily or pericoital step. That is the effectiveness advantage. But the tradeoff is permanence. The counseling has to respect regret, age, reproductive certainty, coercion history, access, and whether the patient understands that reversal is not the plan.

Male sterilization and female sterilization both sit in the less-than-one group, but the procedure, recovery, risk profile, and who bears the bodily burden are different. In real counseling, this matters. A couple asking for permanent contraception should not automatically send the woman to the operating room if vasectomy is acceptable, available, and consistent with their goals.

Now compare sterilization with long-acting reversible contraception.

Long-acting reversible contraception gives a similar actual-use effectiveness advantage while preserving reversibility.

The copper IUD is long-acting and nonhormonal. The FDA chart lists it as inserted and removed by a health care provider, lasting up to ten years. It can be attractive for a patient who wants highly effective contraception without hormones. But the same chart also reminds us that spotting, cramps, and heavier or longer periods can occur.

That matters because "nonhormonal" is not automatically "best."

Imagine a patient with already heavy painful menses who says, I want the most natural method. If the resident simply says copper IUD because it has no hormones, the counseling is incomplete. The method may be effective and nonhormonal, but its bleeding profile may be a poor fit. The right counseling is to explain the tradeoff: highly effective, long-acting, hormone-free contraception, with a possible cost of heavier bleeding and cramping.

Now compare that with the levonorgestrel IUD or IUS.

The levonorgestrel intrauterine method is also in the less-than-one group. It is long-acting, progestin-based, and has strong local endometrial effects. The FDA chart lists menstrual changes, irregular bleeding, and amenorrhea among possible effects.

That is not just a side-effect list. It is physiology.

The levonorgestrel effect on the endometrium is why the same device can be contraception, a bleeding treatment, and a safer estrogen-free option for many patients.

But anatomy still matters.

A patient with heavy bleeding and a normal cavity may be an excellent candidate for a levonorgestrel IUD. A patient with a significantly cavity-distorting submucosal fibroid

is different. The device can be hard to place, may not sit correctly, and may be expelled. So the method family is the same, but the uterus changes the fit.

That is the second clinical pearl:

The device lives in a body, not in a chart.

Consider the implantable rod

It also sits in the less-than-one pregnancy per one hundred women group. It is long-acting systemic progestin. Like intrauterine methods, it removes daily behavior. Unlike intrauterine methods, it does not require a uterine cavity. That distinction becomes useful when the uterus is distorted, when pelvic insertion is unacceptable to the patient, or when the patient wants long-acting contraception without an intrauterine device.

But irregular bleeding can matter. The right counseling is not, this is the most effective, so take it. The right counseling is, this is highly effective and low maintenance, but bleeding unpredictability may be the tradeoff.

Move down the ladder

The contraceptive injection has about four pregnancies per one hundred women in the first year in the FDA chart. It is still a strong method. But now the failure surface is timing.

The drug can work well, but only if the patient returns for injections on schedule. A method that requires a health care encounter every three months is not only a pharmacologic method. It is a transportation method, a clinic-access method, a scheduling method, and sometimes an insurance method.

The chart lists loss of bone density, spotting or bleeding between periods, weight gain, and headache as possible risks or side effects. Lecture 3 will handle DMPA in comorbidity in more detail. For now, remember the family logic: injection is less user-dependent than a daily pill, but more system-dependent than an implant or IUD.

Come the scheduled hormonal methods

Combined pills, extended or continuous combined pills, progestin-only pills, patch, and vaginal ring sit around seven pregnancies per one hundred women in the first year in the FDA chart.

That number surprises some learners because these methods can be highly effective when used perfectly.

But typical use is the clinical reality.

The combined pill and extended or continuous pill require daily swallowing. The progestin-only pill also requires daily use, and some formulations are less forgiving when doses are late. The patch requires a weekly pattern. The ring requires insertion and removal on schedule.

These methods ask the patient to keep a rhythm.

That rhythm may fit some patients beautifully.

A patient who wants cycle control, acne benefit, predictable withdrawal bleeding, or the ability to stop quickly may prefer a scheduled method. A patient who cannot tolerate a device, wants privacy from a partner, or wants to control timing herself may also prefer it.

But the resident has to say the quiet part clearly: scheduled methods are only as good as the schedule the patient can actually live with.

Do not confuse intelligence with adherence. A brilliant patient can miss pills while working nights, caring for children, rotating through hospital shifts, traveling, or hiding medication from an unsafe partner.

The counseling question is not, are you responsible?

The counseling question is, what does your life make easy or hard?

Now consider the contraceptive software application, listed around seven to eight pregnancies per one hundred women in the first year.

This is a good place to teach why "natural" and "low burden" are not synonyms.

App-based or fertility-awareness methods may avoid hormones and devices, but they demand measurement, logging, abstinence or barrier use during fertile windows, and careful interpretation. The FDA chart specifically notes that ovulation prediction may be inaccurate if used when currently or recently on hormonal birth control or hormonal treatments that prevent ovulation.

So for a patient highly motivated to avoid hormones and comfortable with the required behaviors, it may be acceptable. For a patient in whom pregnancy would be dangerous, or whose life makes daily tracking unrealistic, the method may be a poor fit.

Again, the issue is not whether the method is respectable. The issue is whether the failure surface is acceptable.

Move to pericoital and barrier methods

Male condoms are listed at about thirteen pregnancies per one hundred women in the first year. That is less effective for pregnancy prevention than long-acting methods.

But condoms have an additional value that most other methods do not: they can reduce transmission of some sexually transmitted infections.

This creates an important counseling contrast.

If a patient mainly needs pregnancy prevention, condoms alone may not provide enough effectiveness.

If a patient has STI exposure risk, a highly effective contraceptive method alone may not provide the needed infection protection.

So the resident should not say, condoms are worse.

The resident should say, condoms do a different job.

Sometimes the best answer is dual-method counseling: a highly effective contraceptive method for pregnancy prevention plus condoms for infection-risk reduction.

Now the diaphragm, sponge, cervical cap, internal condom, spermicides, and anti-sperm vaginal agents.

These methods sit lower on the pregnancy-prevention ladder because they require correct use around intercourse. The FDA chart lists diaphragm and sponge around seventeen pregnancies per one hundred women, internal condom around twenty one, cervical cap around twenty two to twenty three, and spermicides or anti-sperm vaginal agents around twenty one to twenty eight.

These are not "bad" methods. They are high user-dependence methods.

That distinction matters.

A patient may choose them because she wants no hormones, no device, no procedure, and full control around intercourse. That is a legitimate value structure.

But if pregnancy would be medically dangerous, or if the patient says she often has unplanned sex without supplies available, the resident should not present these methods as equivalent to long-acting contraception.

Counseling is not coercion. But counseling must be numerate.

Move to emergency contraception

Emergency contraception is rescue contraception. It is not routine contraception.

The FDA chart states that levonorgestrel one point five milligrams should be taken as soon as possible within three days after unprotected sex. Ulipristal acetate should be taken as soon as possible within five days.

The most important word is soon.

Emergency contraception is time-sensitive because its benefit is tied to the biology of ovulation and fertilization timing. Waiting turns a rescue method into a weaker rescue.

There is also a common misunderstanding in the chart language.

The chart says levonorgestrel prevents pregnancy in seven out of eight women who would have become pregnant, and ulipristal prevents pregnancy in six or seven out of ten women who would have become pregnant.

That does not mean the pregnancy rate after levonorgestrel is seven out of eight.

It means the opposite. It means that among women who would have become pregnant from that exposure, seven of eight will not become pregnant after using levonorgestrel.

This is a perfect example of why residents must translate numbers, not merely recite them.

A patient hears the number differently from how the chart intends it.

The resident's job is to make the meaning safe.

Step back and ask what the FDA chart is not

It is not a complete eligibility guideline.

It does not tell you whether estrogen is safe in migraine with aura.

It does not tell you how to classify a patient less than twenty one days postpartum.

It does not tell you whether a patient with antiphospholipid antibodies should use combined hormonal contraception.

It does not tell you what to do with graft failure after solid-organ transplant.

That is the work of ACOG's hormonal contraception guidance and the US MEC, which lecture 3 will teach in detail.

But lecture 2 still needs to install the boundary:

The FDA chart answers, how effective is this method group in broad actual use?

The US MEC answers, is this method medically acceptable for this patient?

Those are not the same question.

Let us rehearse with cases

Case one.

A twenty nine year old wants "the best birth control" because pregnancy would be dangerous while she is taking a teratogenic medication.

The tempting error is to hand her the full menu and let her pick without framing risk.

The better counseling is to say: because pregnancy would carry high risk, we should begin with the most effective actual-use methods. That means sterilization if she wants permanent contraception, or long-acting reversible methods such as an IUD or implant if she wants reversibility. Then we discuss whether hormones, insertion, bleeding profile, privacy, and future pregnancy plans change the choice.

The lesson: when pregnancy risk is high, user dependence matters more.

Case two.

A patient wants a hormone-free method and has heavy painful periods.

The tempting answer is copper IUD because it is highly effective and nonhormonal.

The better answer is to explain the tradeoff. Copper is long-acting, highly effective, and hormone-free, but it may worsen bleeding and cramping. A levonorgestrel IUD is hormonal, but mostly local, and may improve bleeding. The patient may still choose copper, but now she is choosing with the relevant physiology in view.

The lesson: "no hormones" is a value, not the whole clinical problem.

Case three.

A patient wants both contraception and treatment for heavy bleeding. Ultrasound suggests a cavity-distorting submucosal fibroid.

The tempting answer is levonorgestrel IUD because it is effective and often improves bleeding.

The better answer is to pause. The method's endometrial effect is useful, but cavity distortion can make placement and retention problematic. This patient may need fibroid-focused evaluation and treatment planning rather than simply method selection.

The lesson: method family must meet anatomy.

Case four.

A patient uses condoms because she is worried about STI exposure, but she also says pregnancy would be unacceptable right now.

The tempting error is to dismiss condoms as low efficacy.

The better counseling is dual: condoms are valuable for reducing some STI risk, but typical-use pregnancy prevention is lower than long-acting methods. If pregnancy prevention is also a high priority, add a more effective contraceptive method and keep condoms for infection-risk reduction.

The lesson: do not make one method do two jobs badly when two methods can do two jobs well.

Case five.

A patient comes after a condom break and asks for the pill that "works seven out of eight times." She thinks that means one in eight women still gets pregnant after levonorgestrel.

The resident should clarify the denominator. The FDA chart is describing women who would have become pregnant from that exposure. Levonorgestrel prevents pregnancy in seven of eight of those expected pregnancies when used within the proper time window. It is rescue, not a replacement for ongoing contraception.

The lesson: emergency contraception numbers must be translated.

Bring the lecture together

Contraceptive counseling has three layers.

Layer one is effectiveness: how many pregnancies occur in actual use?

Layer two is user burden: what does the patient have to do, and how often, for the method to work?

Layer three is fit: does the method match the patient's body, medical risks, reproductive goals, bleeding priorities, privacy needs, infection risks, and tolerance for procedures or hormones?

The FDA chart is excellent for layer one and useful for layer two.

It is not enough for layer three.

That is why the resident should not memorize contraception as a product list.

Memorize the family logic.

Permanent methods remove ongoing user behavior but trade away reversibility.

Long-acting reversible methods remove ongoing adherence while preserving fertility options.

Scheduled hormonal methods can be excellent but depend on rhythm and medical eligibility.

Pericoital methods preserve control around intercourse but carry high user dependence.

Condoms do pregnancy prevention less well than long-acting methods but add infection-risk value.

Emergency contraception is urgent rescue after a contraceptive gap, not an ongoing plan.

The final resident reflex is this:

When a patient asks for the best method, ask best for what.

Then identify the method family, the failure surface, the reversibility goal, the hormone exposure, the bleeding effect, the infection-prevention need, and the medical eligibility boundary.

Efficacy starts the conversation.

Safety, fit, and values choose the method.